Ground and Aerial Systems in Urban Search and Rescue Operations

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ABSTRACT

After natural disasters occur, people can go missing, first responders can not be everywhere at once. This can lead to retrievals being “too late” and victims not being saved in time. With the assistance of ground and aerial systems victim recovery can be expedited with some human assistance…
Add: Why our project is important potentially for a large group of people (not just you)? What are the "pains"/problems that your system is designed to help solve?

NOVELTY

This system functions in both the capacity for victim retreival when their positioning is static or dynamic

REAL-WORLD APPLICATION

Something like this:
Post-Earthquake an aerial asset conducts reconnaissance to find the victim along with traversable terrain and relays the information to the ground asset to assist a trapped victim.

FEATURES

Map sharing

Target Sharing

Pursuit Algorithms

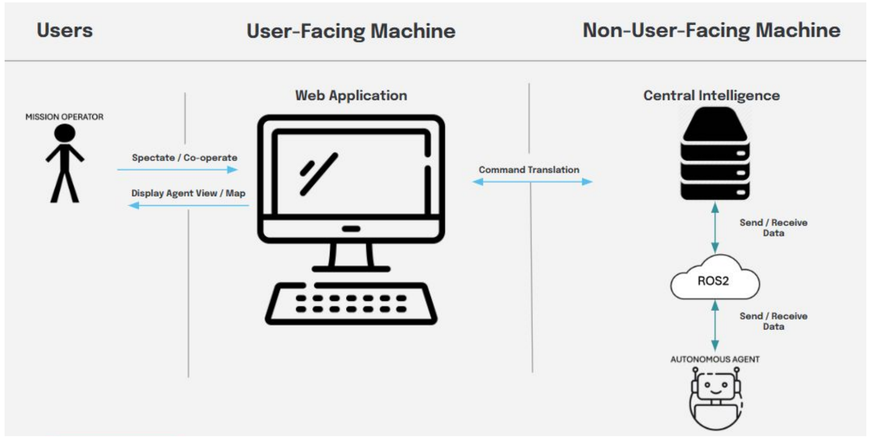
ACCOMPLISHMENTS

Breakthroughs during the milestones

What our project can currently do

What we are most proud of

DESIGN/SYSTEM ARCHITECTURE



Diagram

Image of Robots in action

Web App Behind the scenes

Captions with each image

Tools

FUTURE DEVELOPMENTS

Stretch goals

Performance metrics and how to improve

NORTHROP GRUMMAN

ENGINEERING & SCIENCE
STUDENT DESIGN SHOWCASE

FLORIDA TECH

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